

Math 196 Problem Set 7

Due May 14, at 10:00 PM

Problem 1. Determine which of the following matrices are orthogonal.

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \frac{1}{7} \begin{bmatrix} 6 & -2 & 3 \\ 2 & -3 & -6 \\ -3 & 6 & -2 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

Problem 2. Consider the rotation matrix

$$R = \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}$$

Find an invertible matrix S such that $S^{-1}RS$ is not an orthogonal matrix.

Problem 3. Show that if $u_1, \dots, u_n$ and $v_1, \dots, v_n$ are orthonormal bases, then the unique matrix A such that $Au_i = v_i$ for each i is orthogonal.

Problem 4. For each of the following linear systems, $Ax = v$ find a least squares approximate solution $\tilde{x}$, and determine the distance $|A\tilde{x} - v|$.

(a) $\begin{bmatrix} 1 \\ 3 \\ -2 \\ 4 \end{bmatrix} x = \begin{bmatrix} -2 \\ -3 \\ -1 \\ 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} x = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 \\ 2 & 8 \\ 1 & 5 \end{bmatrix} x = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & -1 \\ 1 & 2 \\ -1 & 0 \end{bmatrix} x = \begin{bmatrix} -5 \\ 1 \\ 3 \end{bmatrix}$

For the next three problems, you should set up a linear system by hand. You can use software, for example https://wims.univ-cotedazur.fr/wims/en_tool-linear-linsolver.en.html, to find the solution. You are encouraged to plot the data points and best fit functions using graphing software, for example <https://www.desmos.com/calculator> and <https://www.desmos.com/3d>, but you do not need to submit these plots.

Problem 5. Using the method of least squares, find the line $y = ax + b$ that best fits the following points (x, y) :

$$(0, 3), (1, 3), (2, 7), (3, 8)$$

Problem 6. Using the method of least squares, find the function of the form $z = ax + by + c$ that best fits the following points (x, y, z) :

$$(0, 0, -1), (1, 2, 1), (-1, 1, 0), (2, 0, 3), (0, -1, -1)$$

Problem 7. Using the method of least squares, find the function of the form $y = vt - gt^2/2$ that best fits the data points (t, y) :

$$(0, 0), (1, 7), (2, 7), (3, 2), (4, -8)$$

Problem 8. Suppose that we have some data points $(x_1, y_1), \dots, (x_N, y_N)$, and we think that y does not depend on x , so $y = c$, an unknown constant. If we use the method of least squares to fit a function of this form to the data, find an interpretation of the constant we get.¹

Problem 9. For which values of t is the matrix

$$A(t) = \begin{bmatrix} t & 2t \\ 3t & t^2 + 2t \end{bmatrix}$$

invertible? For values of t where $A(t)$ is not invertible, find a basis for its kernel.

Problem 10. Consider the following matrix.

$$A = \begin{bmatrix} 2/3 & -4/3 & -4/3 \\ -4/3 & 2/3 & -4/3 \\ -4/3 & -4/3 & 2/3 \end{bmatrix}$$

This represents a transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$.

(a) Describe the effect of T on the vector

$$v = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

(b) Describe the effect of T on vectors in $\text{span}\{v\}^\perp$.

(c) Find a basis for $\mathbb{R}^3$ consisting of v and two vectors in $\text{span}\{v\}^\perp$, and find the matrix B for the transformation T with respect to this basis. Write down an equation of the form $A = SBS^{-1}$.

¹. It may be helpful to start by considering the cases of $N = 2$ or $N = 3$ if you are finding it difficult to reason about arbitrary N .